

When AR HUDs Become Distractors: Resource-Dependent Memory-Driven Attentional Capture and Design Implications

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ABSTRACT

Virtual and augmented reality (VR/AR) systems frequently employ head-up displays (HUDs) to enhance interaction efficiency, yet immersive environments impose substantial demands on working memory (WM) and attention. Information maintained in WM can trigger memory-driven attentional capture (MDAC), diverting attention from external events and increasing the risk of attentional tunneling. This work examines the resource-dependent boundary of MDAC and proposes an XR paradigm to test its generalization to immersive contexts. Behavioral experiments on 2D displays showed MDAC-related performance costs were amplified under high-demand search relative to low-demand search. These findings suggest that in visually complex HUDs, actively maintained WM content can more strongly influence attention, highlighting the need to consider cognitive status alongside display characteristics in immersive interface design.

Index Terms: memory-driven attentional capture, visual working memory, augmented reality, head-up displays

1 INTRODUCTION

Virtual and augmented reality (VR/AR) systems frequently integrate head-up displays (HUDs) to reduce eyes-off-road time and support efficient interaction by projecting information directly within the user's primary field of view [1, 2]. While these designs aim to enhance usability, immersive VR environments impose substantially higher attentional and working memory demands than traditional 2D laboratory displays. Prior work has shown that expanded fields of view, high visual complexity, and stereoscopic depth processing increase cognitive load. Consequently, users in XR often operate under elevated baseline attentional load, limiting the availability of resources for top-down control and suppression of task-irrelevant stimuli.

HMD-based stereoscopic viewing additionally requires resolving binocular disparity, while persistent peripheral stimulation in wide field of view environments demands sustained attentional regulation. Within this context, HUD-provided information (e.g., colors, symbols, icons, or system states) frequently serves as reference information that must be actively maintained in WM to guide navigation and decision-making. Cognitive research demonstrates that such internally maintained WM content can bias attentional selection toward matching stimuli in the environment, a phenomenon known as memory-driven attentional capture (MDAC) [3, 4]. While adaptive in some contexts, this bias can increase susceptibility to cognitive capture or attentional tunneling in immersive environments.

Previous HUD design research has emphasized visual clutter, display density, and spatial layout. In contrast, comparatively little attention has been paid to how internally maintained WM content interacts with visual selection during HUD use. This gap is

particularly consequential in immersive XR environments, where attentional resources are already heavily taxed by wide fields of view, depth processing, and complex scene integration. Under these conditions, HUD elements can act as an attentional attractor, drawing limited processing resources toward WM-matching information even when it is task-irrelevant. Understanding the boundary conditions of this WM-based bias is therefore crucial for assessing hidden cognitive costs and ensuring user safety in immersive environments.

2 BACKGROUND AND MOTIVATION

Classical Feature Integration Theory distinguishes parallel visual search, which relies on distinct diagnostic features that "pop-out", from serial search, which requires focused, item-by-item inspection under high target-distractor similarity [5]. Cognitive research on MDAC shows that information actively maintained in WM can bias these search processes. Behavioral evidence from traditional 2D displays indicates that MDAC-related performance costs are most robust under high attentional filtering demand (i.e., serial search), when efficient feature-based selection is insufficient [6].

Immersive XR environments differ fundamentally from 2D laboratory displays. Wide visual fields, constant peripheral stimulation, and stereoscopic depth processing collectively increase demands on attentional control mechanisms, reducing the efficacy of top-down suppression. This raises a central question: does WM-driven attentional capture operate similarly in immersive contexts, or does XR amplify the influence of WM content by reducing the availability of attentional control resources?

3 COMPLETED BEHAVIORAL STUDY

Prior to XR extension, two laboratory experiments were conducted using conventional computer displays. In both experiments, participants maintained a visual feature in WM while performing a visual search task. During search, a task-irrelevant singleton distractor was either absent, present but non-matching, or present and matching the maintained WM feature.

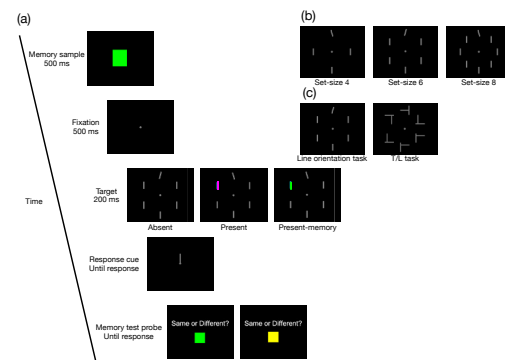


Figure 1: Example trial sequence of Experiments 1 and 2 with singleton distractor (a), set sizes (b), and task paradigm (c).

- **Experiment 1: Line search (Low filtering demand).** Distinct line orientations supported efficient, largely parallel processing [7, 8]. Reaction time (RT) analyses revealed a significant main

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effect of distractor presence, $F(2, 54) = 8.50$, $p < .001$, with reliable slowing for WM-matching distractors. A set size $\times$ presence interaction, $F(4, 108) = 3.12$, $p = .018$, indicated that WM-related slowing with increased with set size.

- **Experiment 2: T/L search (High filtering demand).** In contrast to Experiment 1, this task required spatial feature integration to resolve high target–distractor similarity [7, 8]. RT analyses showed a larger main effect of distractor presence, $F(2, 56) = 16.23$, $p < .001$, with WM-matching distractors producing more robust slowing than in the low-demand condition. No significant set size $\times$ presence interaction was observed, $F(4, 112) = 1.65$, $p = .166$.

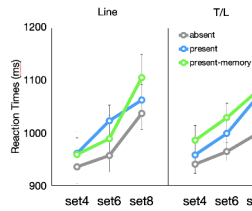


Figure 2: Mean reaction times for Experiment 1 (line search) and Experiment 2 (T/L search). Error bars represent ± 1 SEM.

Takeaway: MDAC interference is resource-dependent and more robust under high attentional filtering demand.

Across experiments, a three-way interaction between task type, set size, and distractor presence demonstrated that WM-driven interference is resource-dependent, becoming substantially more pronounced under higher filtering demand.

4 XR EXTENSION: PROPOSED HUD PARADIGM

The boundary conditions identified from laboratory experiments are directly relevant to XR HUD design. Immersive interfaces combine visually rich environments with persistent overlays maintained in WM. These conditions closely resemble the conditions under which MDAC produced robust behavioral costs in the laboratory. To assess whether WM-driven attentional capture generalizes to immersive contexts, we propose a controlled XR paradigm that translates core attentional manipulations into a realistic HUD-based task.



Figure 3: Proposed XR experimental setup

Participants will view a straight-road driving scene from a first-person, head-stabilized perspective in stereoscopic VR. Each trial begins with a brief (500 ms) presentation of a colored directional arrow that must be actively maintained in WM. Following this, participants search for a specific target vehicle and identify its turn signal direction (Left/Right). To ensure WM maintenance, a probe task follows, requiring a match/non-match judgment on the arrow's color.

Search demand is manipulated via morphological similarity between the target and distractors (Fig 4). In the **low-demand (parallel)** condition, the target (angular SUV) and distractors (round SUVs) have high shape contrast, allowing the target to "pop-out" via feature-based guidance. In the **high-demand (serial)** condition, the target and distractors share high structural similarity (e.g., subtle grill or headlight differences), requiring item-by-item inspection.

Orthogonal to search demand, HUD includes three distractor conditions: absent, present (non-matching), and present-memory (WM-matching). To prevent ceiling performance in the low-demand condition and ensure comparability across conditions, task difficulty is equated using an adaptive staircase procedure [9]. This procedure dynamically adjusts the luminance of peripheral distractors (e.g., vehicle turn signals), maintaining mean accuracy at approximately 75–85%. This design allows isolation of how

WM-maintained HUD information disrupts visual selection in XR, and whether such disruption is selectively amplified under higher attentional demand.

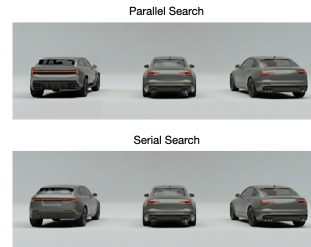


Figure 4: XR search examples across low and high-demand conditions.

5 DISCUSSION AND IMPLICATIONS

The present work aimed to characterize how MDAC may operate in immersive XR environments by combining controlled behavioral evidence with a HUD-based design. Our laboratory experiments demonstrated that MDAC is resource-dependent, with more robust interference under high attentional filtering demand than under low demand. Extending this to XR allows testing whether MDAC persists when baseline attentional resources are already taxed. If the proposed XR experiments reveal an attenuated difference between low- and high-demand search, this would suggest that immersion increases susceptibility to WM-based attentional bias by limiting top-down control.

A primary contribution of this work is shifting the HUD evaluation from physical complexity (e.g., display density) to the cognitive status of HUD information. From an applied perspective, this highlights an underexplored risk in XR HUD design: under high visual or attentional demand, WM-maintained HUD elements may automatically capture processing resources, delaying responses to behaviorally critical events. Effective XR interface design should therefore consider not only the visual properties of overlays but also whether HUD information is likely to be actively maintained in WM, and how such maintenance may increase susceptibility to cognitive capture.

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